

Program : Diploma in Renewable Energy	
Course Code : 6299B	Course Title: Microprocessor Lab
Semester : 6	Credits: 1.5
Course Category: Programme Elective	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- Develop and execute assembly language programs for solving arithmetic and logical problems using microprocessor

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Microprocessors and programming		Digital Electronics & Microprocessor	5

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the addressing modes and decoding of 8085 microprocessor	9	Applying
CO2	Develop and execute assembly language programs using Arithmetic instructions	9	Applying
CO3	Develop and execute assembly language programs using branch instruction	9	Applying
CO4	Develop and execute assembly language programs to interface converters with 8085	12	Applying
	Lab Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Demonstrate the addressing modes and decoding of 8085 microprocessor		
M1.01	Demonstrate the parts of a microprocessor trainer kit and the decoding techniques	1	Understanding
M1.02	Make use of different <u>addressing modes for</u> data transfer	2	Applying
M1.03	Develop and execute ALP for block data transfer	3	Applying
M1.04	Develop and execute ALP for block of data interchange	3	Applying
CO2	Develop and execute assembly language programs using Arithmetic instructions		
M2.01	Develop and execute ALP for Addition of two 8 bit and 16 bit numbers with and without carry	3	Applying
M2.02	Develop and execute ALP for Subtraction of two 8 bit numbers with and without borrow	3	Applying
M2.03	Develop and execute ALP for Multiplication and division of two 8 bit numbers	3	Applying
	Lab Test – I	3	

CO3	Develop and execute assembly language programs using branch instruction		
M3.01	Develop and execute ALP to find largest and smallest number in a data array	3	Applying
M3.02	Develop and execute ALP for Sorting of arrays in ascending and descending order.	3	Applying
M3.03	Develop and execute ALP for BCD to Binary conversion and vice versa	3	Applying
C04	Develop and execute assembly language programs to interface 8085 with peripherals		
M4.01	Develop ALP to Interface 8085 with various displays	3	Applying
M4.02	Develop ALP to interface DAC/ADC with 8085	3	Applying
M4.03	Develop ALP for stepper motor/ DC motor interfacing using 8085	3	Applying
M4.04	Develop ALP for Traffic light interfacing using 8085	3	Applying
	Lab Test – II	3	

Text / Reference

T/R	Book Title/Author
T1	Ramesh Gaonkar, Microprocessor Architecture Programming and Applications, Penram International Publishing; Sixth edition, 2014.
T2	Mohamed Ali Mazidi, Janice Gillispie Mazidi, "The 8051 microcontroller and embedded systems using Assembly and C", second edition, Pearson/Prentice hall of India.
R3	Kenneth. J. Ayala, The 8051 microcontroller, 3rd edition, Cengage Learning, 2010

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/108107029
2	https://www.slideshare.net/kpaulraj/microprocessor-lab
3	https://www.youtube.com/watch?v=liRPtvj7bFU
4	https://www.iitk.ac.in/new/microprocessor-and-microcontroller-laboratory
5	https://learning-microprocessors.sourceforge.io/8085-simulator.html

Student Activity

Suggested Open-ended Experiments:

Students can do open ended experiments as a group of 3-5. There is no duplication in experiments in between groups. This is mainly for the purpose of continuous internal evaluation. Students should prepare a separate report on open ended experiment of their choice.

Examples.

1. Develop and execute ALP to find square and square root of a number
2. Develop and execute ALP to find ASCII equivalent of key pressed
3. Develop and execute ALP to find factorial of a number
4. Develop and execute ALP to convert ASCII to Binary and vice versa
5. Develop and execute ALP to search smaller number in a data array and find its square